

ISAAC BANKART

LINKEDIN

EDUCATION

M.SC. Electrical and Information Technology — Since 10/2023

Karlsruhe Institute of Technology — Karlsruhe

- Specialisation: Information Theory and Telecommunications Engineering
- Lab courses: Communications Engineering Lab
- Master's thesis: "Toward Fault-Tolerant Quantum Error Correction"
- current grade: 1,7 (1="very good", 2="good")

B.SC. Electrical and Information Technology — 10/2019–09/2023

Karlsruhe Institute of Technology — Karlsruhe

- Lab courses: Hardware and Software for Power Electronic Systems, Filter Design with Digital Signal Processing
- Bachelor's Thesis: "Optimization of Gallager B Decoder over Binary Symmetric Channels"
- Final grade: 2,2

A-Level (Abitur) Sixth Form — 09/2013–08/2018

Prince William School — Northamptonshire England

- A-Levels: Physics, Maths, Product Design (Abitur 1,3)

WORK EXPERIENCE

Research Student, Communications Engineering Lab — Since 09/2026

Toward Fault-Tolerant Quantum Error Correction — Karlsruhe

- Realising a quantum quaternary Relay-BP Decoder

Research Student, Communications Engineering Lab — 03/2024–09/2025

Channel Coding on the Binary Symmetric Channel — Karlsruhe

- Matlab, PyTorch, neural networks and machine learning

Internship, SEW Eurodrive Forschungscampus — 04/2022–04/2023

Implementing new Communicationsystems — Bruchsal

- Configuring C/C++ Click Modular Router in Linux
- Multimodal Channel Measurement and Packet Disection

LEISURE ACTIVITIES

Music Production — Since 08/2020

- Independent Recording and Production/Mixing across various genres on spotify:

  including singer/songwriter, electronic, and fasion design,

Software Projects

- Microcontroller programming and implementing algorithms in JavaScript, Bash, Python, C/C++ (github)

Building and Construction

- House renovation and maintainance work at home and with friends.